DSDG Hackathon — Model 1, 2 & 3 Documentation

Nov 7, 2026 · @Eshaan

Overview

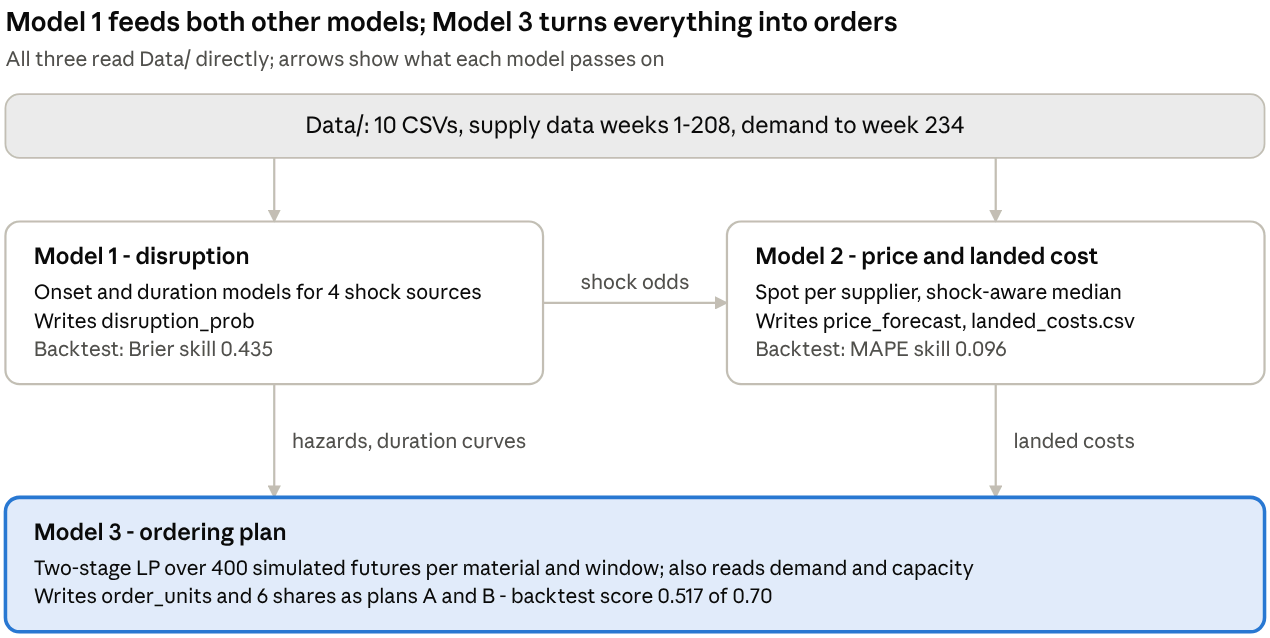
Three Python modules fill the submission's nine columns, each feeding the next; Model 3's ordering columns carry 70% of the score.

Score part	Weight	Column(s)	Built by	Compared against
Disruption forecast (Brier skill)	18%	`disruption_prob`	Model 1	each material's historical disruption rate
Price forecast (MAPE skill)	12%	`price_forecast`	Model 2	carrying the last benchmark price forward
Financial F	40%	`order_units`, `share_KES` to `share_AUR`	Model 3	naive policy and a perfect-hindsight plan
Fill rate in shock weeks	21%	same	Model 3	share of demand met in shock weeks + 4 recovery weeks
Diversification	9%	shares	Model 3	1 - HHI of shares, even split = 1

Model	Code	Run notebook	Repo folder	Writes
1 Disruption	`model1.py`	`model1_run.ipynb`	`Code Base/pipelining/Model 1/iter3`	`disruption_prob`, hazards, duration curves
2 Price	`model2.py`	`model2_run.ipynb`	`Code Base/pipelining/Model 2/iter3`	`price_forecast`, `landed_costs.csv`

Model	Code	Run notebook	Repo folder	Writes
3 Ordering	`model3.py`	`model3_run.ipynb`	`Code Base/pipelining/Model 3`	`order_units`, shares, `submission_A.csv`, `submission_B.csv`

All three share one team file, `Model 1/iter3/outputs/submission.csv`. Each notebook fills only its own columns, so they run in order: 1, then 2, then 3.



model pipeline - 3 models, 1 shared submission file

Model 2 refits Model 1 internally and Model 3 refits both, so all three always use the same forecast week and the same data.

Data findings

The data follows clean rules: four shock sources rebuild the `disrupted` column exactly (2,080 of 2,080 rows), and each source has its own warning signal, length and cost effect. All three models are built around these sources.

Source	How it is detected	Past events (weeks 1-208)	Length	Warning signal	Effect
Single supplier	capacity below 70% of normal while its lane is open	7	2-11 weeks	that supplier's futures premium ramps up 1-4 weeks ahead; above 0.06 it preceded a shock 13 of 13 times	all 3 suppliers' spot +15-35% while it lasts
Country	2+ suppliers of one country hit in the same week	1 (Kestria, weeks 99-133)	35 weeks	tension, fx depreciation, PMI drop	capacity 20% of normal; spot +5-20%
Lane closure	transit days at least 1.5x normal	4 (Atlantic once, Merrow 3 times)	2-8 weeks	congestion, about 1 week ahead; above 50 it preceded a closure 19 of 19 times	capacity 75-80%, lead time +3 to +4 weeks, container rate 1.6-3x; spot barely moves
Tariff	tariff above the supplier's base rate	3	10-18 weeks	announced 2-8 weeks ahead in `tariffs.csv`	landed cost only

Other findings that shaped the models:

- The 31 material-level episodes are only 15 real events. Counting episodes doubles the odds a shock lasts 13+ weeks (0.26 vs 0.13).
- No signal warns more than about 4 weeks ahead. Three of the 15 shocks had no warning at all; news reported them in their first week.
- In calm weeks futures predict nothing: they lose to carry-forward when the premium is under 3%.
- The benchmark price is exactly the mean of the 3 supplier spot prices.
- Landed cost = spot x (1 + tariff) + b x container rate + c per supplier, with b = 0 on the NORTH, SOUTH and RAIL lanes. Average error 0.06%.
- For 9 of 10 materials the cheapest supplier ships through the Merrow Strait (Kestria for 7, Sunda Coast for 2): the cheapest route is also the riskiest.
- Demand is known through week 234, so all uncertainty is on the supply side.
- The naive policy in `sample_submission.csv` orders each week's demand, 100% from the supplier with the largest normal capacity (Kestria for 7 materials).

Model 1: disruption forecast

Model 1 forecasts the probability that each material is disrupted in each future week by combining four source-level forecasts. On 26 past forecast weeks it scores a Brier skill of 0.435 against the base rate.

How it works

1. **Rebuild the sources** from `supply_history`, `tariffs` and `shipping` for every supplier, country and lane, every week.
2. **Source active now:** a duration curve gives P(still active h weeks ahead) = S(age + h) / S(age). Curves are discrete Weibull, one per source type, fit on the 15 de-duplicated events. Type curves are pulled toward a shared curve, and 15% of a long tail (mean 40 weeks) is mixed in for a shock longer than anything in training.
3. **Source calm now:** a logistic model predicts P(first onset within k weeks). Each k gets its own base rate; signal effects differ by lead bucket (1, 2, 3, 4, 5-8, 9+ weeks). The weekly increments are combined with the duration curve to give P(active at week h).
4. **Announced tariffs** are scheduled: active from the effective week with probability 0.95, then the tariff duration curve.
5. **Combine:** P(material disrupted) = 1 - product of (1 - p) over its sources, each lane and country counted once, clipped to [0.02, 0.98].

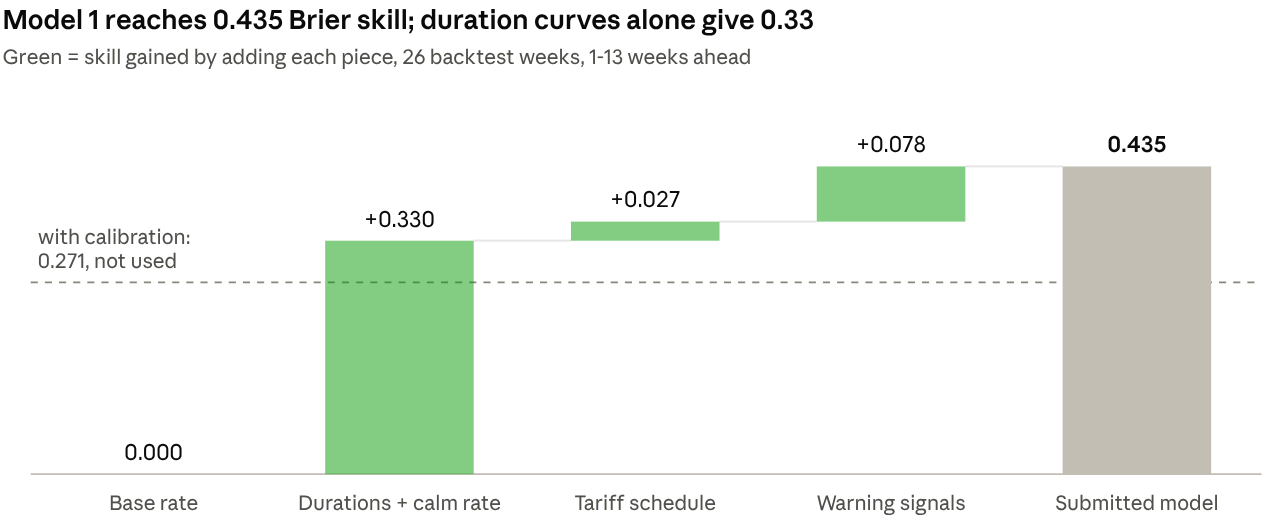
Onset model	Features (all computed from data up to the forecast week)
Single supplier	futures premium above 0.02, 2-week premium change, country tension level, 3-week change, z-score
Country	tension level, 3-week change, z-score, fx 4-week change, PMI 4-week change
Lane	congestion above 38, 1- and 2-week change, z-score

Left out on purpose: transit days, container rate, capacity and lead time (they move only when a shock starts, so they flag a shock but never warn of one), and material name (would memorize which materials got hit).

Changes from the team plan

- **Supplier-level sources, not one material-level model.** This matches how the data is generated and gives Model 3 its hazards directly.
- **Country events split out.** Before the split, tension spikes from one-supplier shocks pushed every supplier in that country up (lithium reached 0.98 with nothing behind it).
- **"Within k weeks" labels** instead of "exactly k weeks". Skill at the doc's four test weeks rose from 0.37 to 0.71.
- **Platt calibration off.** Fit on earlier periods it cut skill from 0.435 to 0.27; too few events to fit it.
- **No gradient boosting.** Regularization strength barely changed results; the limit is the number of events.
- **26 backtest origins** (every 4 weeks from week 104) instead of 4, because the doc's 4 origins contain only 4 new shocks.

Results



Model 1 backtest, 26 origins (weeks 104-204), Brier skill vs each material's base rate

Knowing whether a shock is on and how long shocks last does most of the work; the warning signals add the rest, mostly on calm weeks (0.10 to 0.29).

Skill fades with lead time: 0.88 at 1 week, about 0.65 at 2-4 weeks, about 0.13 past 9 weeks. Calibration is close to the diagonal below 0.2 and slightly underconfident between 0.25 and 0.45. Most remaining error comes from shock endings (Kestria ended at week 133, Merrow at 180) and no-warning shocks.

Outputs

Output	Used by
`disruption_prob` (both windows)	submission; Model 2's jump rule
`hazards.csv`: per supplier and week, P(active) and P(starts that week) for supplier, country, lane and tariff sources	Model 3 scenario generator
`hazards_sources.csv`: the same for lanes and countries	Model 3 (one draw per lane or country, applied to every supplier on it)
`duration_curves.csv`: P(shock lasts at least x weeks) by type	Model 3 shock lengths

Model 2: price forecast and landed costs

Model 2 forecasts each supplier's spot price and averages the three into the benchmark. It cuts price error from 6.08% to 5.49% (MAPE skill 0.096) and landed-cost error from 6.4% to 5.6% versus carry-forward. It refits Model 1 internally, so the two stay consistent.

How it works

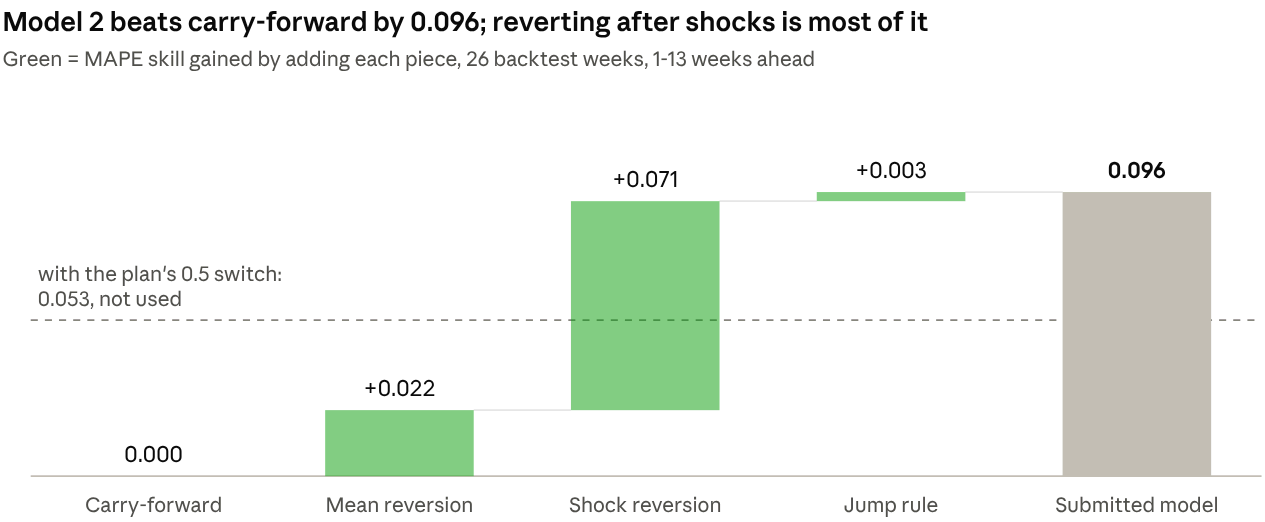
1. **Calm material:** carry-forward plus drift, minus a pull back toward its 26-week calm average. The pull grows with horizon (0.06 at 1 week to 0.52 at 13 weeks). Both are fit per horizon by least absolute error.
2. **Material in a shock now:** two scenarios, today's elevated price and the pre-shock price, weighted by Model 1's chance the shock is still on. The forecast is the median of that mix, including normal price noise.
3. **Calm, but a shock is coming:** scenarios are the calm forecast and the calm forecast plus the typical onset jump (median about 16%), only up to 4 weeks ahead, where Model 1's signals reach.
4. **Landed cost** per supplier and week = forecast spot x (1 + scheduled tariff) + b x forecast container rate + c, in three versions (below).

Changes from the team plan

- **Futures baseline dropped.** It scored -0.007 (worse than carry-forward) because futures carry no information in calm weeks.

- **The 0.5 jump switch replaced** by the median of the shock/no-shock mix. The hard switch flips all-or-nothing at the wrong times (0.053 vs 0.096).
- **Jumps limited to 4 weeks ahead.** Further out they only added calm-week errors.
- **No boosted trees.** Not needed to beat carry-forward; this column is 12% of the score.

Results



Model 2 backtest, 26 origins (weeks 104-204), MAPE skill vs carry-forward

The team plan's futures baseline (not shown) scored -0.007, below carry-forward.

Gains come from shock periods (skill 0.20 when a shock is active at the forecast week) and long horizons (0.13 at 9-13 weeks); calm weeks 1-4 weeks out are close to carry-forward, as they should be. Copper, rare earths, natural gas and cobalt gain 17-20%. Lithium is negative (-0.11) and steel flat: both Kestria materials, hurt early in the 35-week Kestria event when Model 1 expected it to end soon.

Outputs

Column in `landed_costs.csv`	Meaning	Use it for
`landed_cost_expected`	weighted by Model 1's shock odds	Model 3's quick LP
`landed_cost_calm`	no shock at all: calm price, base tariff, normal container rate	Model 3's scenario LP, which adds its own shock costs

Column in `landed_costs.csv`	Meaning	Use it for
`landed_cost`	single most likely path, consistent with `price_forecast`	reporting
`spot_*`, `tariff_*`, `container_*`	the pieces	rebuilding cost under other scenarios

`price_forecast` goes into the team file for both windows.

Model 3: ordering plan

Model 3 picks one set of orders per material and window that works well across 400 simulated futures. On 10 test windows plan A averages 0.517 of a possible 0.70, against 0.175 for the naive policy and 0.656 for a perfect-hindsight plan.

How it works

1. **Simulator** of the competition rules:
   - start with 5 weeks of demand in stock
   - each country's share is filled up to its capacity that week; unfilled units are cancelled and not charged
   - filled units are paid at that week's landed cost and arrive after that week's lead time
   - holding costs 2% and each unmet unit 5x the average landed cost per week
   - stock and units in transit at the end sell back at 70%
2. **Naive policy** inferred from `sample_submission.csv`: order this week's demand, 100% from the largest-capacity supplier.
3. **Two-stage LP** per material and window (`scipy` HiGHS, interior point). Orders are shared across futures; fills, stock and shortfall are per future. Fills are capped by orders and capacity, each supplier gets at least a minimum share of each week's order, and the objective is the weighted average cost across futures. The internal stockout penalty is lambda x average cost. Futures that are identical for a material are merged with their weights added (one window: 13 s to 1 s).

4. **Scenario generator** from Model 1: for each lane, country, supplier and tariff, the start week is drawn from Model 1's onset probabilities and the length from its duration curves. Capacity cut, extra lead time and price jump are copied from a random past episode of the same type. The same draws are used for all 10 materials, so one Merrow closure hits every Merrow supplier, and shocks active at the forecast week carry into every future.
5. **Stress futures** with total weight omega: Merrow closed 8 weeks, each material's cheapest supplier banned, all capacity halved.
6. **Post-processing** per the doc: whole-number `order_units`, shares from orders, 0 for countries that don't sell the material, even split when the total is 0.

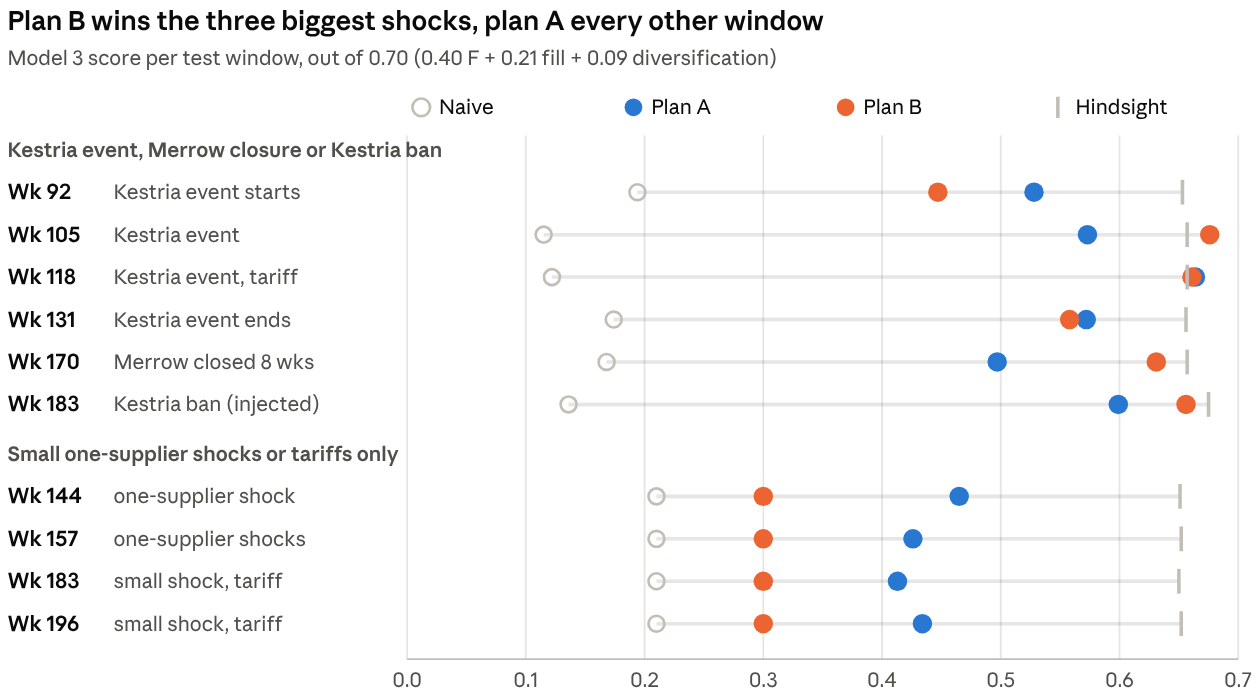
Scoring used locally

- **F** = (naive cost - our cost) / (naive cost - hindsight cost), summed over materials, clipped to [0, 1].
- **Fill** = mean over materials of demand met in shock weeks + 4 recovery weeks (all weeks if the window had no shock).
- **Diversification** = mean over rows of (1 - HHI of shares) / (1 - 1/3).
- **Score** = 0.40 F + 0.21 fill + 0.09 diversification, out of 0.70.

Test windows start at weeks 92, 105, 118, 131, 144, 157, 170, 183 and 196, plus window 183 with an 8-week Kestria ban injected into the actuals. Each is planned from the week before it, with no later data (checked: plans are identical when later data is removed).

Results

Plan	Mean score	Worst window	Cost vs naive	Fill in shocks	Diversification
Perfect hindsight (ceiling)	0.656	0.650	-39%	100%	0.51
Plan A	**0.517**	**0.413**	-28%	97.5%	0.77
Plan B	0.483	0.300	-24%	100%	1.00
Quick LP (one forecast + safety buffer)	0.413	0.222	-14%	88%	0.67
Naive	0.175	0.115	0%	83%	0



Model 3 backtest, 9 past windows + 1 injected Kestria ban, scored locally

In the three biggest shocks plan B scores close to the perfect-hindsight plan; everywhere else plan A is ahead, by 0.13-0.17 when shocks are small.

Plans A and B

Kaggle keeps the better of two final picks, so the two plans bet on different futures.

- **Plan A** (stress weight 0.05, minimum share 0.1, lambda 5, Model 1 probabilities as is, 400 futures) wins calm windows. It was picked from a 37-setting grid by worst window; the top settings were within seed noise, and 200 futures were seed-sensitive, so it uses 400.
- **Plan B** (stress weight 0.15, minimum share 0.33, lambda 8, probabilities x1.5) never ran out of stock in testing and wins big shocks: 0.656 vs 0.599 under the Kestria ban, 0.631 vs 0.497 in the Merrow-closure window. In calm windows it costs about 5% more than naive, so F is 0 there.
- **Proposed update to A:** since B covers shocks, A can specialize in calm windows (stress weight 0, minimum share 0.05, probabilities x0.5). The better of A and B then averages 0.575 instead of 0.546, and A's calm-window score rises from 0.435 to 0.529.

Public leaderboard: plan B scored 44.15. That fits a calm public window: about 30 points from B's ordering columns (fill 21 + diversification 9 + F 0) plus about 14 from Models 1 and 2. This is an inference; Kaggle shows only the total.

How to run

Run the three notebooks in order in Databricks; each finds the data folder and the other models' code on its own, and the last one leaves an upload-ready file.

1. **Model 1** (`Model 1/iter3/model1_run.ipynb`): checks the labels rebuild exactly, fills `disruption_prob` (creating the team file from the sample the first time), and writes the hazard files and an ablation chart.
2. **Model 2** (`Model 2/iter3/model2_run.ipynb`): checks the futures and landed-cost assumptions, fills `price_forecast` , and writes `landed_costs.csv` .
3. **Model 3** (`Model 3/model3_run.ipynb`): backtests naive, quick LP, plan A and plan B (a few minutes), writes plan A into the team file and both plans to `outputs/submission_A.csv` and `submission_B.csv` , and runs the upload check on every column.

Each notebook prints READY TO UPLOAD or a list of what to fix. Slow cells (Model 1 ablation, Model 3 tuning) load saved results unless their `RERUN_` flag is set to True. If you edit a `.py` file, restart the kernel or `importlib.reload` it.

When Phase 2 data lands

1. Put the new weeks in `Data/` under the same file names. If the organizers send only the new weeks, append them with the commented cell in Model 1's last section.
2. Run the three notebooks again in order. Each one's last cell detects the new data, re-plans the private window (weeks 222-234) from week 221, and re-scores.
3. In Model 1's last cell, check `signal_extremes` : any signal past its training maximum is the earliest warning of the big shock.
4. Upload the new `submission_A.csv` and `submission_B.csv` .

Final submission

- Select **both** plan A and plan B as the two final picks; Kaggle keeps the better one.
- Deadline from the team plan: Sunday 12 PM Central (Nov 8, 2026).
- If Phase 2 data arrives too late to re-run, select the current A and B. Their weeks 222-234 are planned from week 208.

Assumptions, weak spots and open items

The backtest numbers are optimistic: every structural choice was made after studying all 208 weeks, so expect lower real scores.

Assumptions to confirm against the official rules (all are settings in `model3.Config`):

- How F combines materials (here: total cost over all 10).
- Which weeks count as "shock + 4 recovery", and what counts when a window has no shock.
- How diversification is averaged (here: plain mean over rows).
- What "average landed cost" means for holding, stockout and sellback (here: mean over the material's suppliers and window weeks).
- Settled: the two readings of "5 weeks of demand in stock" change naive's cost by 0.04%.

Known weak spots:

- Model 1's country model learned from one event (Kestria), so it overreacts to tension spikes; at week 172 it gave Kestria up to 50% within 13 weeks.
- Model 2 is negative for lithium and flat for steel, from the early weeks of the Kestria event.
- Shock endings are inherently unpredictable and cause most remaining error in Models 1 and 2.
- Three of 15 past shocks had no warning; only the 0.02 probability floor and Model 3's stress futures cover them.
- A shock bigger than anything in training is covered by Model 1's long-tail duration mix, Model 3's stress futures and plan B, but none of these can be tested against a real one.

Fixed during testing: Model 3's normal-capacity figures briefly used weeks after the forecast week; they now use only earlier weeks, and plans are identical with later data removed.

Open items:

- [ ] Get plan A's public score: about 55-61 expected if the scoring reading is right.
- [ ] Decide whether to switch plan A to the calm-specialist settings.
- [ ] Find the official evaluation text to confirm the scoring assumptions.
- [ ] Re-run all three notebooks after Phase 2 data lands, then select A and B as finals.